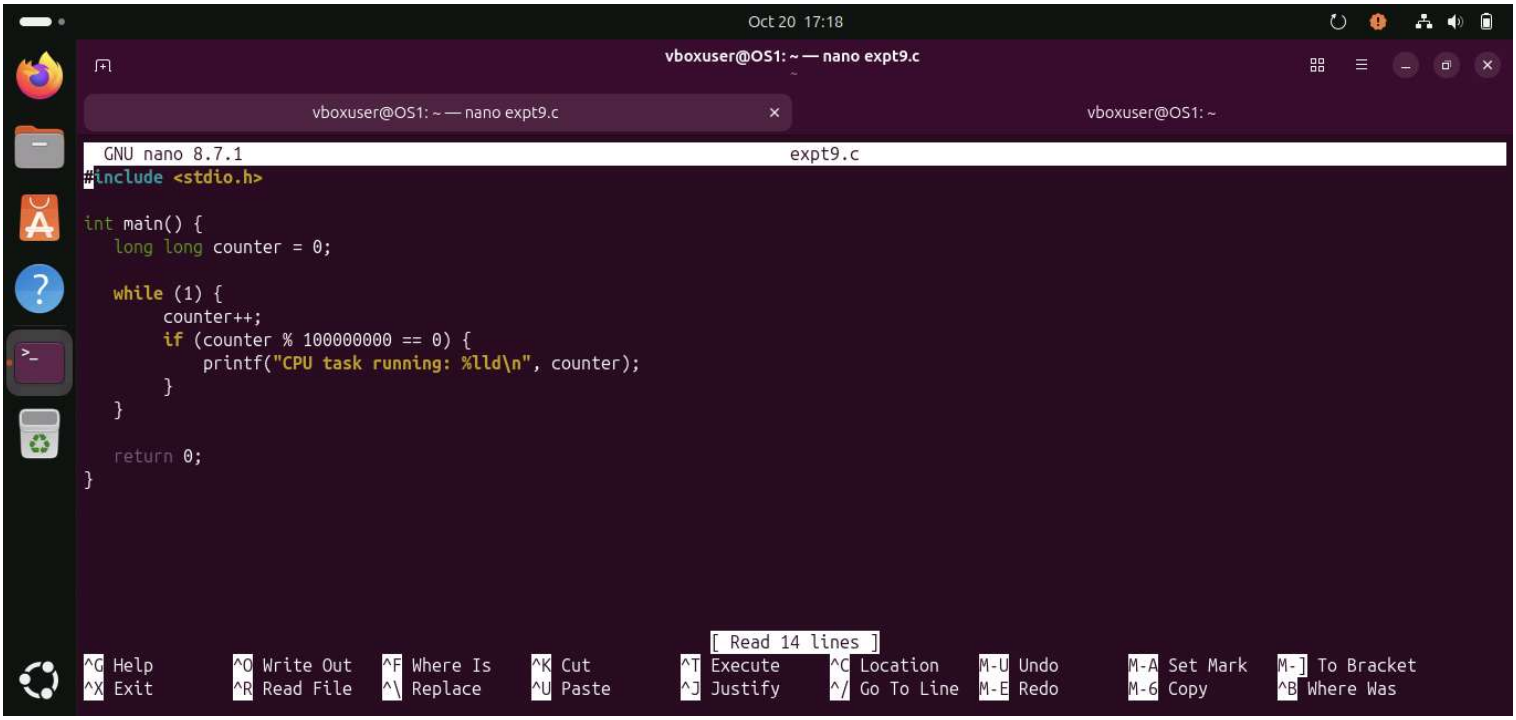
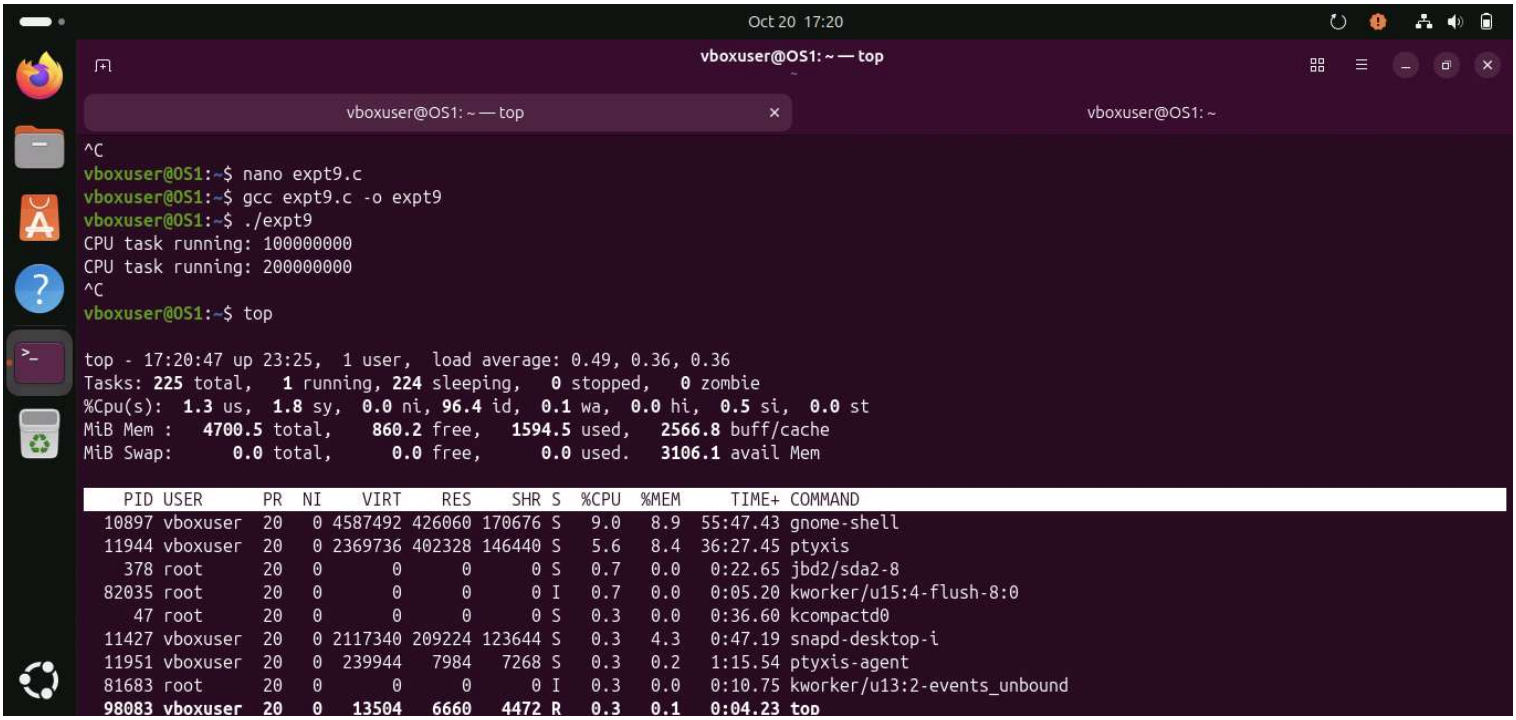
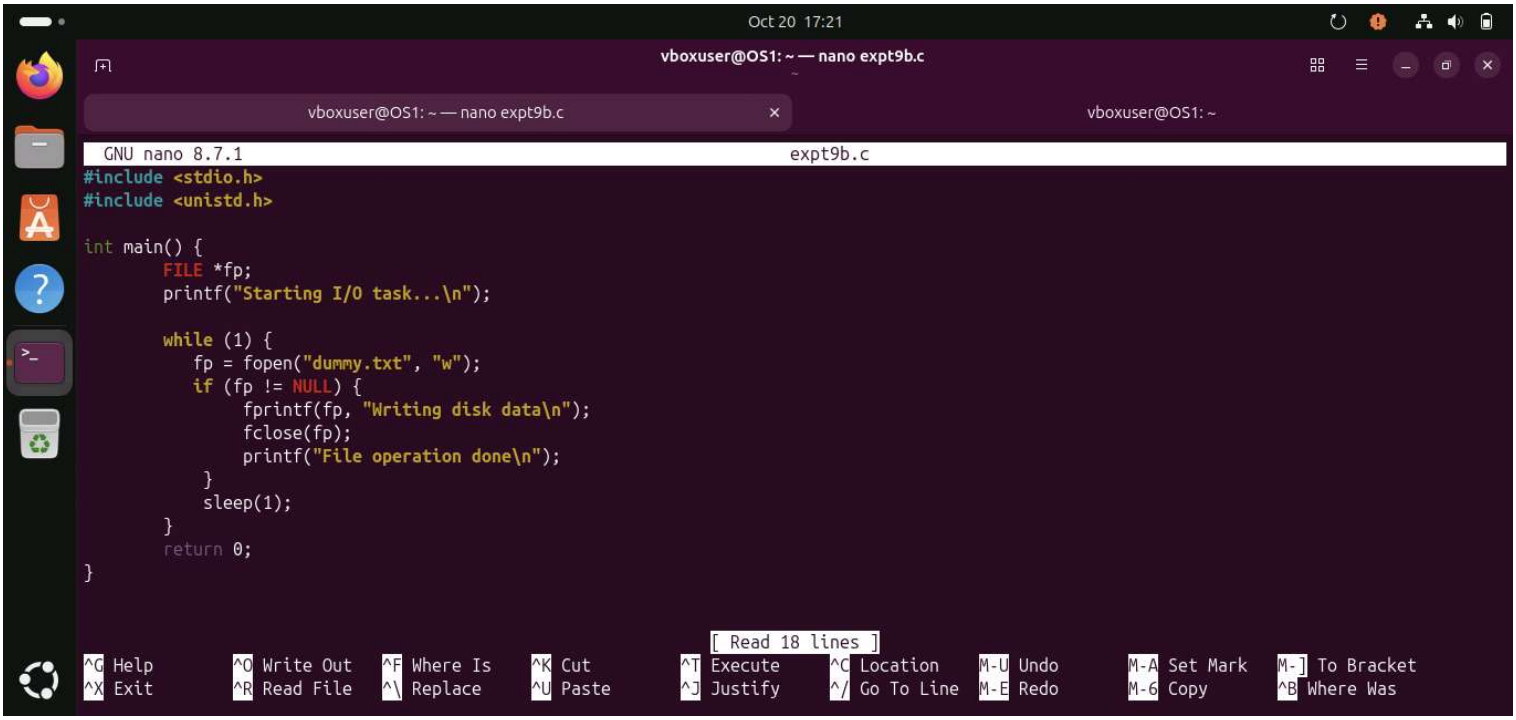

Oct 20 17:18
vboxuser@OS1: ~ — nano expt9.c
vboxuser@OS1: ~ — nano expt9.c
vboxuser@OS1: ~
GNU nano 8.7.1 expt9.c
#include <stdio.h>
int main() {
long long counter = 0;
while (1) {
counter++;
if (counter % 100000000 == 0) {
printf("CPU task running: %lld\n", counter);
}
}
return 0;
}
[Read 14 lines]
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^B Where Was


Oct 20 17:20
vboxuser@OS1: ~ — top
vboxuser@OS1: ~ — top
vboxuser@OS1: ~
^C
vboxuser@OS1:~$ nano expt9.c
vboxuser@OS1:~$ gcc expt9.c -o expt9
vboxuser@OS1:~$ ./expt9
CPU task running: 100000000
CPU task running: 200000000
^C
vboxuser@OS1:~$ top
top - 17:20:47 up 23:25, 1 user, load average: 0.49, 0.36, 0.36
Tasks: 225 total, 1 running, 224 sleeping, 0 stopped, 0 zombie
%Cpu(s): 1.3 us, 1.8 sy, 0.0 ni, 96.4 id, 0.1 wa, 0.0 hi, 0.5 si, 0.0 st
MiB Mem : 4700.5 total, 860.2 free, 1594.5 used, 2566.8 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 3106.1 avail Mem
PID USER PR NI VIRT RES SHR S %CPU %MEM TIME+ COMMAND
10897 vboxuser 20 0 4587492 426060 170676 S 9.0 8.9 55:47.43 gnome-shell
11944 vboxuser 20 0 2369736 402328 146440 S 5.6 8.4 36:27.45 ptyxis
378 root 20 0 0 0 0 S 0.7 0.0 0:22.65 jbd2/sda2-8
82035 root 20 0 0 0 0 I 0.7 0.0 0:05.20 kworker/u15:4-flush-8:0
47 root 20 0 0 0 0 S 0.3 0.0 0:36.60 kcompactd0
11427 vboxuser 20 0 2117340 209224 123644 S 0.3 4.3 0:47.19 snapd-desktop-i
11951 vboxuser 20 0 239944 7984 7268 S 0.3 0.2 1:15.54 ptyxis-agent
81683 root 20 0 0 0 0 I 0.3 0.0 0:10.75 kworker/u13:2-events_unbound
98083 vboxuser 20 0 13504 6660 4472 R 0.3 0.1 0:04.23 top


Oct 20 17:21
vboxuser@OS1: ~ — nano expt9b.c
vboxuser@OS1: ~ — nano expt9b.c
vboxuser@OS1: ~
GNU nano 8.7.1
expt9b.c
#include <stdio.h>
#include <unistd.h>
int main() {
FILE *fp;
printf("Starting I/O task...\n");
while (1) {
fp = fopen("dummy.txt", "w");
if (fp != NULL) {
fprintf(fp, "Writing disk data\n");
fclose(fp);
printf("File operation done\n");
}
sleep(1);
}
return 0;
}
[Read 18 lines]
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^B Where Was